

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics
Year 8

EXTENSION WORKBOOK | E1–E2 SELF-STUDY

Probability

Year 8 Mathematics • Extension workbook

Name _____ Class _____ Teacher _____

How this workbook works

Each lesson is **three pages**. Copy your teacher's **notes**, then work through **We do** (dark boxes: we solve these together; copy every step) and **You do** (your turn, one round at a time). Rate each success criterion at the end.

Your notes and examples match the main workbook, so you follow the class lesson, then do these harder questions: page 1 **Medium** ●●, page 2 **Hard** ●●●, page 3 **Going further**. E1 and E2 are **self-study**: their key ideas are printed and their examples are worked for you.

NSW Mathematics K–10 Syllabus (2022), Stage 4: MA4-PRO-C-01; extension (Stage 5 Path): MA5-PRO-P-01. We acknowledge the Bundjalung people, the Traditional Custodians of the land on which we learn.

Contents

1	Chance and the sample space	2
2	Probability of an event	5
3	Complementary events	8
4	Probability problems	11
5	Observed probability	14
6	Expected frequency	17
E1	Venn diagrams (Stage 5 Path)	20
E2	Two-way tables (Stage 5 Path)	23
+	At the back: chapter summary and if you were away 26, review 29, exit tickets 31, homework strips 35	

Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

1 Chance and the sample space

Probability • MA4-PRO-C-01 **MEDIUM ••**

LEARNING INTENTION

We are learning to describe chance with words and with numbers from 0 to 1, list a sample space, and decide whether outcomes are equally likely.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* place an event on the probability scale from 0 to 1. ○○○
- *I can* list the sample space of a chance experiment. ○○○
- *I can* decide, with a reason, whether outcomes are equally likely. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Answer each part. Where it says explain, give a reason.

a The scale

Write each letter where its event belongs on the scale. A: the sun rises tomorrow. B: rolling a 7 on a die. C: a coin lands heads.



b Sample space

The spinner is spun once. List the sample space, then count the outcomes.



c Equally likely?

The spinner is spun once. Is each colour equally likely? Explain.



d Sample space

List the sample space for the day of the week a person was born on. How many outcomes are there?

YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER Chance words: impossible (0) · unlikely (below $\frac{1}{2}$) · even chance ($\frac{1}{2}$) · likely (above $\frac{1}{2}$) · certain (1)

ROUND 1 Questions 1–4: each spinner is spun once. Count the blue sectors out of all the sectors, then write the chance word for landing on blue.

1 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

2 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

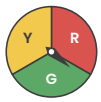
3 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

4 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

ROUND 2 Questions 5–8: list the sample space inside braces { } and count the outcomes. Write repeated outcomes only once.

5 This spinner is spun once. List the sample space.



Sample space { _____ }

Number of outcomes _____

6 One tile is picked. List the sample space.



Sample space { _____ }

Number of outcomes _____

7 List the possible last digits of a car number plate.

Sample space { _____ }

Number of outcomes _____

8 3 coins are tossed together. List the possible numbers of heads.



Sample space { _____ }

Number of outcomes _____

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER List each outcome once. An event is the outcomes you want. Compare events by counting their outcomes. A probability is never below 0 or above 1.

ROUND 3 Questions 9–11: list the outcomes in each event, inside braces { }.

9 A 10-sided die numbered 1 to 10 is rolled. List the outcomes in the event 'a prime number'.



Outcomes { }

10 A month is picked. List the outcomes in the event 'the month has exactly 30 days'.

Outcomes { }

11 A normal die is rolled. List the outcomes in the event 'a factor of 12'.



Outcomes { }

ROUND 4 Questions 12–14: decide if the outcomes are equally likely. Write *yes* or *no*, then give a reason.

12 3 coins are tossed. Are 0, 1, 2 and 3 heads equally likely?



Yes or no

Reason

13 A person is picked. Is every letter equally likely to be the first letter of their first name?

Yes or no

Reason

14 A random number generator makes a whole number from 1 to 20. Is each number equally likely?



Yes or no

Reason

ROUND 5 Questions 15–17: put the events in order from least likely to most likely. Explain using the number of outcomes in each event.

15 A die is rolled. Order these events: A an even number, B 5 or more, C less than 2.



Order

Reason

16 One tile is picked. Order these events: (i) a B, (ii) an I, (iii) a vowel.



Order

Reason

17 The spinner is spun once. Order these events: A red, B blue, C yellow.



Order

Reason

ROUND 6 Questions 18–20: use probabilities as numbers from 0 to 1. Show how you know.

18 Write a decimal that could be the probability of an event that is unlikely but not impossible.

Answer

19 One event has probability 0.45 and another has probability $\frac{4}{9}$. Which event is more likely? Show your working.

Answer

20 Explain why a probability can never be 1.2 or −0.3.

Answer

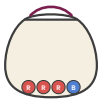
1 Chance and the sample space

GOING FURTHER

WE DO with your teacher: copy every step into the space

Example 2 Answer each part in a full sentence, and give the reason.

a Equally likely?
One counter is taken from the bag without looking. List the sample space. Are the outcomes equally likely?



b A match
Kingscliff plays Cabarita in a final. List the sample space of results. Are the outcomes equally likely?

c Order
A die is rolled. Put these in order from least likely to most likely: A an even number, B a 6, C a number less than 6, D a 9.



d Forecast
The forecast is a 30% chance of rain. Is rain likely or unlikely? Could it still rain?



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER Equally likely means the same chance, not just the same number of names. Likely means more than half; unlikely means less than half, but more than 0.

1 Colour these 5 equal sectors so that red is twice as likely as blue, and green is impossible. Label each sector with its colour.



Answer

2 Spinner A has 3 equal sectors, 1 of them blue. Spinner B has 8 equal sectors, 3 of them blue. Which spinner gives the better chance of blue? Explain.

Answer

3 A bag holds 10 counters, all red or blue. Red is likely. Blue is unlikely but possible. How many red counters could there be? List every possibility.

Answer

4 Jo writes the sample space for this spinner as {red, red, blue}. What is wrong? Write the sample space correctly.



Answer

5 Year 7

Put these probabilities in order from least likely to most likely: A 0.6, B $\frac{3}{5}$, C 62%, D $\frac{5}{8}$.

Answer

6 A die is weighted so that a 6 comes up half the time. Are the six outcomes equally likely? Write the chance word for rolling a 6.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 The surf club picks 1 of its 12 members at random: 5 are in Year 8, 4 in Year 9 and 3 in Year 10. Write the chance word for 'the person is in Year 8' and for 'the person is in Year 11'. Give reasons.

Answer

2 A spinner has 8 equal sectors coloured red, blue and yellow. Red is likely. Blue and yellow are equally likely, and yellow is possible. Find every possible design (how many sectors of each colour).

Answer

3 In a game a die is rolled: Ava wins on 1 or 2, Ben on 3, 4 or 5, and Cal on 6. Is the game fair? Explain, then change the rules to make it fair.

Answer

2 Probability of an event

Probability • MA4-PRO-C-01

MEDIUM ••

LEARNING INTENTION

We are learning to find the probability of an event when the outcomes are equally likely, and write it as a fraction, decimal and percentage.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* count the favourable outcomes and the total outcomes. ○○○
- *I can* use P(E) = favourable ÷ total, and simplify. ○○○
- *I can* write a probability as a fraction, a decimal and a percentage. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Write it as a fraction in simplest form.

a Spinner

The spinner is spun once. Find the probability it lands on red, P(red).



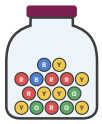
b Die

A die is rolled once. Find the probability of an odd number, P(odd).



c Lollies

One lolly is taken without looking. Find P(yellow or blue).



d Tiles

One tile is picked without looking. Find the probability of a vowel, P(vowel).

K	I	N	G	S	C	L	I	F	F

YOU DO on your own: work down the box, one round at a time. Show your working. **8 QUESTIONS**

REMEMBER Probability = $\frac{\text{favourable outcomes}}{\text{total outcomes}}$. Simplify: divide the top and bottom by the same number.

ROUND 1 Questions 1–4: each spinner is spun once. Find P(blue) as a fraction in simplest form. If the sectors are not equal, compare their sizes.

1 Find the probability of landing on blue. Simplify.



Working _____

P(blue) = _____

2 Find the probability of landing on blue. Simplify.



Working _____

P(blue) = _____

3 Find the probability of landing on blue. Simplify.



Working _____

P(blue) = _____

4 Find the probability of landing on blue. Simplify.



Working _____

P(blue) = _____

ROUND 2 Questions 5–8: a 10-sided die numbered 1 to 10 is rolled once. List the favourable outcomes, then find each probability as a decimal.

5 Find the probability of a prime number.



Favourable outcomes { _____

Probability _____

6 Find the probability of a multiple of 3.



Favourable outcomes { _____

Probability _____

7 Find the probability of a factor of 10.



Favourable outcomes { _____

Probability _____

8 Find the probability of at least 8 (8 or more).



Favourable outcomes { _____

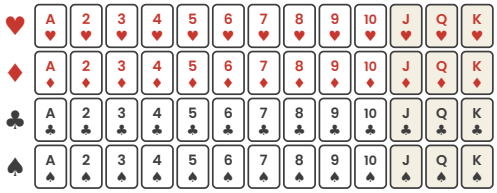
Probability _____

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER 'Or' with no overlap: add the counts. Decimal: top ÷ bottom. Percentage: × 100%. Missing total = count ÷ probability.

ROUND 3

Questions 9–11: one card is taken from a standard deck without looking. Find each probability as a fraction in simplest form.



J, Q, K are picture cards (shaded). An ace is not a number card here.

9 Find the probability of a black picture card.

Working _____

Probability _____

10 Find the probability of a card with an even number on it (2, 4, 6, 8 or 10).

Working _____

Probability _____

11 Find the probability of a card from 2 to 5 (including 2 and 5).

Working _____

Probability _____

ROUND 4

Questions 12–14: one coin falls out of the money box from Example 2. Write each answer in the form asked.



20 coins in the money box.

12 Find the probability of a 10c or 20c coin. Write it as a percentage.

Working _____

Answer _____

13 Find the probability of a coin worth less than $1. Write it as a percentage.

Working _____

Answer _____

14 Find the probability of a coin worth at least 50c (50c or more). Write it as a decimal.

Working _____

Answer _____

ROUND 5

Questions 15–17: work backwards from the probability to find the missing number. Show your working.

15 A bag has 6 red counters and some blue counters. $P(\text{red}) = \frac{1}{3}$. How many blue counters are in the bag?

Answer _____

16 A spinner has 12 equal sectors. $P(\text{green}) = 0.25$. How many sectors are green?

Answer _____

17 8K has 25 students. For a student picked at random, $P(\text{plays a sport}) = \frac{3}{5}$. How many students in 8K play a sport?

Answer _____

ROUND 6

Questions 18–20: find each probability. Write it as a fraction, a decimal and a percentage. Round decimals to 2 places if needed.

18 One tile is picked from P R O B A B I L I T Y. Find the probability of a vowel.

Fraction _____

Decimal _____

Percentage _____

19 A whole number from 1 to 40 is chosen at random. Find the probability of a multiple of 8.

Fraction _____

Decimal _____

Percentage _____

20 A month of the year is chosen at random. Find the probability that it starts with J.

Fraction _____

Decimal _____

Percentage _____

WE DO with your teacher: copy every step into the space

Example 2 Give each answer in the form the question asks for: fraction, decimal or percentage.

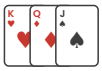
a As a decimal

One coin falls out of the money box. Find the probability it is a gold coin ($1 or $2). Write it as a decimal.



b Cards

One card is taken from a 52-card deck. Find the probability it is a red picture card (J, Q or K).



c As a percentage

A die is rolled. Find the probability of a number greater than 4. Write it as a percentage.



d Notes

Elena has two $5, four $10 and three $20 notes. She takes one without looking. Find the probability it is a $20 note.

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER Probability = favourable ÷ total. If the total changes, the probability changes. Never count an outcome twice.

1 An Esky has 8 lemonade, 5 orange and 2 lime cans. How many cola cans must be added so that $P(\text{cola}) = \frac{2}{5}$?

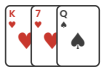


Answer

2 Start again with the Esky from Question 1 (no colas). How many cola cans must be added so that $P(\text{orange}) = 0.25$?

Answer

3 One card is taken from a standard deck. Find the probability that it is a heart or a picture card. Careful: do not count any card twice.



Answer

4 One tile is picked without looking. Find the probability that it is an L or an M.



Answer

5 The spinner's sector angles are 180° red, 90° blue, 60° green and 30° yellow. Find P(green) and P(yellow). (A full turn is 360°.)



Answer

6 A bag holds 40 marbles and $P(\text{red}) = 0.35$. How many red marbles are in the bag? How many marbles are not red?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 The digits 1, 3, 5 and 8 are each used once to make a 4-digit number at random. Find the probability that the number (a) starts with 5 (b) is odd (c) is more than 5000.

Answer

2 A jar holds 60 lollies. $P(\text{red}) = \frac{1}{4}$ and $P(\text{green}) = \frac{1}{3}$. The rest are yellow. How many yellow lollies are there? Then find P(yellow).

Answer

3 Kai picks a whole number from 1 to 20 at random. Kai wins if it is a multiple of 3 or a multiple of 4. Find the probability that Kai wins.

Answer

3 Complementary events

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LEARNING INTENTION

We are learning to identify the complement of an event, and use $P(\text{not E}) = 1 - P(E)$ to solve problems.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* write the complement of an event. ○○○
- ▸ *I can* show that $P(E) + P(\text{not E}) = 1$. ○○○
- ▸ *I can* use $P(\text{not E}) = 1 - P(E)$ with fractions, decimals and percentages. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Write the complement (the 'not' event), then find the probabilities.

a Complement

A die is rolled. Write the complement of 'rolling a 6' and list its outcomes.



b Cards

A pack has 10 red, 6 yellow and 4 green cards. One card is taken without looking. Find P(yellow) and P(not yellow).

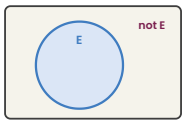
c Decimal

The probability of rain is 0.15. Find the probability of no rain.



d Diagram

A die is rolled. E = 'an even number'. Write the outcomes in the diagram, then find P(not E).



YOU DO on your own: work down the box, one round at a time. Show your working. **8 QUESTIONS**

REMEMBER The complement must include every other outcome. $P(E) + P(\text{not E}) = 1$.

ROUND 1 Questions 1–4: write the complement of each event. Make sure it includes every other possible outcome.

1. Write the complement of 'winning a hockey game'.

   Complement _____

2. Write the complement of 'being born in winter'.

   Complement _____

3. Write the complement of 'being under 15 years old'.

   Complement _____

4. A card is drawn from a deck. Write the complement of 'a heart'.

   Complement _____

ROUND 2 Questions 5–8: a fair die is rolled once. Find P(E) and P(not E) for each event E.

5. E is 'rolling at least 2 (2 or more)'. Find P(E) and P(not E).

   P(E) = _____
   P(not E) = _____

6. E is 'rolling at most 4 (4 or less)'. Find P(E) and P(not E).

   P(E) = _____
   P(not E) = _____

7. E is 'rolling a multiple of 3'. Find P(E) and P(not E).

   P(E) = _____
   P(not E) = _____

8. E is 'rolling a number greater than 6'. Find P(E) and P(not E).

   P(E) = _____
   P(not E) = _____

3 Complementary events

Probability • MA4-PRO-C-01

HARD •••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER $P(\text{not E}) = 1 - P(\text{E})$. All the probabilities add to 1. 'Not A and not B' means every outcome except A and B.

ROUND 3 — Questions 9–11: each probability is given. Write the probability that the event does NOT happen.

9 $P(\text{choosing a jack from a deck}) = \frac{1}{13}$. Find $P(\text{not choosing a jack})$.

Answer _____

10 $P(\text{shooting a goal}) = 61\%$. Find $P(\text{missing})$.

Answer _____

11 $P(\text{winning a prize}) = 0.15$. Find $P(\text{not winning a prize})$.

Answer _____

ROUND 4 — Questions 12–14: a jar has 40 red, 25 blue, 50 black and 15 white jelly beans. One is taken without looking. Find each probability in simplest form.

12 Find the probability that it is not yellow.

Working _____

Answer _____

13 Find the probability that it is not red.

Working _____

Answer _____

14 Find the probability that it is not black and not white.

Working _____

Answer _____

ROUND 5 — Questions 15–17: the probabilities of all the outcomes add to 1. Find the missing probability.

15 Tomorrow will be sunny, cloudy or rainy. $P(\text{sunny}) = 0.45$ and $P(\text{cloudy}) = 0.3$. Find $P(\text{rainy})$.

Working _____

Answer _____

16 A spinner is red, blue or green. $P(\text{red}) = \frac{1}{3}$ and $P(\text{blue}) = \frac{1}{4}$. Find $P(\text{green})$.

Working _____

Answer _____

17 A netball team can win, draw or lose. $P(\text{win}) = 55\%$ and $P(\text{draw}) = 5\%$. Find $P(\text{lose})$.

Working _____

Answer _____

ROUND 6 — Questions 18–20: one tile is picked from P R O B A B I L I T Y without looking. Find each probability.

P	R	O	B	A	B	I	L	I	T	Y

18 Find the probability that it is not a P.

Working _____

Answer _____

19 Find the probability that it is not a vowel.

Working _____

Answer _____

20 Find the probability that it is not an A and not a B.

Working _____

Answer _____

WE DO with your teacher: copy every step into the space

Example 2 Use P(not E) = 1 - P(E). Give each answer in the same form as the question.

a Percentage
A golfer sinks 74% of putts. Find the probability that the golfer misses a putt.

b Missing value
A bag holds only red, blue and white toy cars. P(red) = 0.4 and P(white) = 0.25. Find P(blue).

c Choose
Which is the complement of 'winning a race'? A coming last B coming second C not winning D coming 2nd or 3rd

d Raffle
1000 raffle tickets are sold and Tahnee buys 5. Find the probability that Tahnee does not win.



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER P(E) + P(not E) = 1. When one probability is a multiple of another, write an equation.

1. A car park has 450 cars, 100 motorbikes and 10 buses. One vehicle is picked at random. Which is P(not a car)?
   A $\frac{45}{56}$ B $\frac{110}{450}$ C $\frac{10}{560}$ D $\frac{11}{56}$

   Answer

2. A spinner is only red, blue or green. P(red) is twice P(blue), and P(green) = 0.1. Find P(red).

   Answer

3. The probability that E does not happen is three times the probability that it does. Find P(E).

   Answer

4. A bag holds 50 counters. P(not red) = 0.64. How many red counters are in the bag?

   Answer

5. A 2-digit number (10 to 99) is picked at random. Find the probability that it is not a multiple of 10.

   Answer

6. Can P(E) = 0.7 and P(not E) = 0.4 for the same event? Explain.

   Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

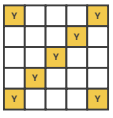
1. A bag of 40 toy cars holds only red, blue and white cars. P(red) = 0.4 and P(white) = 0.25. How many cars of each colour are there? What is the probability of a pink car?

   Answer

2. A raffle sells 800 tickets for one prize. Jai buys some tickets. The probability that Jai does not win is 0.985. How many tickets did Jai buy?

   Answer

3. The grid game has 7 yellow squares out of 25. How many more squares must be coloured yellow so that P(not winning) = 0.6?



   Answer

4 Probability problems

Probability • MA4-PRO-C-01

MEDIUM ●●

LEARNING INTENTION

We are learning to choose and use the right probability skill to solve problems from lists, tables and words.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* decide which skill a problem needs. ○○○
- *I can* set out working with P(E) = favourable ÷ total. ○○○
- *I can* explain my answer in a sentence. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Use the checklist in your notes. Write each probability as a fraction in simplest form, unless the question asks for a percentage.

a A table

28 students on a trip were born in these countries. One is picked at random. Find P(born in Asia).

Australia	17
Italy	3
Vietnam	4
Japan	1
China	3

b Letters

One tile is picked without looking. List the sample space. Is each letter equally likely? Explain.

S U C C E S S

c Percentage

The spinner is spun once. Find the probability of an even number. Write it as a percentage.



d Not

Alex takes one sock without looking from 2 black, 2 blue and 2 red socks. Find the probability that it is not red.



YOU DO on your own: work down the box, one round at a time. Show your working. **8 QUESTIONS**

REMEMBER Probability = favourable outcomes ÷ total outcomes. Write fractions in simplest form. For a decimal, divide the top by the bottom.

ROUND 1 Questions 1–4: one ball is picked at random from a barrel of 100 balls numbered 1 to 100. Find each probability as a fraction in simplest form.

1 Find the probability of a number that is at most 85 (85 or less).

Working _____
Probability _____

2 Find the probability of a multiple of 7.

Working _____
Probability _____

3 Find the probability of a two-digit number.

Working _____
Probability _____

4 Find the probability that the number has a 5 as one of its digits.

Working _____
Probability _____

ROUND 2 Questions 5–8: an 8-sided die has 2 red, 3 white, 1 blue and 2 yellow faces. It is rolled once. Find each probability as a decimal.

5 Find the probability of blue.



Working _____
Probability _____

6 Find the probability of yellow.



Working _____
Probability _____

7 Find the probability of not red.



Working _____
Probability _____

8 Find the probability of white or yellow.



Working _____
Probability _____

4 Probability problems

Probability • MA4-PRO-C-01

HARD •••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER List the outcomes carefully. Count each object, not each name. With ‘or’, never count an outcome twice.

ROUND 3

Questions 9–11: a tennis tournament has 34 players from NSW, 20 from Victoria and 6 from Queensland. One player is picked at random. Find each probability.

9 Find the probability that the player is from Queensland.

Working _____

Answer _____

10 Find the probability that the player is not from Victoria.

Working _____

Answer _____

11 Find the probability that the player is from Queensland or Victoria.

Working _____

Answer _____

ROUND 4

Questions 12–14: a whole number from 1 to 15 is chosen at random. List the favourable outcomes, then find the probability.

12 Find the probability of a number that is even or a multiple of 5 (count each number once).

Favourable outcomes { } _____

Probability _____

13 Find the probability of a number that is not a factor of 12.

Favourable outcomes { } _____

Probability _____

14 Find the probability of a number that is both prime and odd.

Favourable outcomes { } _____

Probability _____

ROUND 5

Questions 15–17: the 10 letters of KINGSCLIFF are written on 10 cards. One card is picked without looking.

K	I	N	G	S	C	L	I	F	F

15 Is each different letter equally likely to be picked? Explain.

Working _____

Answer _____

16 Find the probability that the letter is also in the word FISH.

Working _____

Answer _____

17 Find the probability that the letter is not a vowel.

Working _____

Answer _____

ROUND 6

Questions 18–20: a truck carries 240 boxes of hand sanitiser, 305 of soap, 335 of paper and 120 of pens. One box is picked at random. Find each probability in simplest form.

18 Find the probability that it is a box of hand sanitiser.

Working _____

Answer _____

19 Find the probability that it is a box of pens or soap.

Working _____

Answer _____

20 Find the probability that it is not a box of paper.

Working _____

Answer _____

4 Probability problems

GOING FURTHER

WE DO with your teacher: copy every step into the space

Example 2 Harder problems: read each one twice, decide what you need to find, then solve.

a Small chances

A charity raffle sells 100 000 tickets and Sophia buys 4. Find the probability that Sophia wins. Write it as a decimal.



b Or

A whole number from 1 to 15 is chosen at random. Find the probability of a multiple of 3 or a multiple of 5.



c How many?

A bag of lollies has P(red) = 0.5 and P(green) = 0.2. The rest are purple. Find P(purple). Then find how many of each colour are in a bag of 10 lollies.

d No memory

A coin lands tails 7 times in a row. Find the probability of tails on the next toss.



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER Plan before you calculate. A game is fair when every player has the same probability of winning.

1 Bag A has 10 lollies and bag B has 100 lollies. In both bags, P(red) = 0.5. Which bag gives the better chance of taking a red lolly? Explain.

Answer

2 A whole number from 1 to 50 is chosen at random. Find the probability of a multiple of 4 or a multiple of 6 (count each number once).

Answer

3 A 3-digit number from 100 to 999 is chosen at random. Find the probability that all three digits are the same.

Answer

4 A die is weighted so that P(6) = 0.25. The other five numbers are equally likely. Find P(1).

Answer

5 In a class, the probability that a student picked at random is left-handed is 0.15. There are 3 left-handed students. How many students are in the class?

Answer

6 Ali says: 'A game has two outcomes, win or lose, so P(win) = $\frac{1}{2}$.' Explain what is wrong with Ali's thinking.

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 The canteen sold 120 lunches: 45 wraps, 30 sushi and the rest burgers. One receipt is picked at random. (a) Find P(burger). (b) 20 more burgers are sold. Find the new P(burger).

Answer

2 One card is picked from cards numbered 1 to 20. Ash wins on a multiple of 4, Bo wins on a prime number, and Cy wins on any other number. Is the game fair? Who is most likely to win?

Answer

3 A bag holds 20 marbles and P(red) = 0.35. How many red marbles are there? How many blue marbles must be added to make P(red) = 0.25?

Answer

5 Observed probability

Probability • MA4-PRO-C-01 **MEDIUM ••**

LEARNING INTENTION

We are learning to find observed probabilities from experiments and data, and compare them with theoretical probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find an observed probability: frequency ÷ total trials. ○○○
- *I can* tell theoretical and observed probability apart. ○○○
- *I can* run a simulation with a random number generator. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Observed probability = the number of times it happened ÷ the total number of trials.

a Observed

A die is rolled 60 times and a 2 comes up 15 times. Find the observed probability of a 2.



b Compare

Find the theoretical probability of a 2 on a fair die. Is the observed probability in part a higher or lower?

c Spinner

Vinh spins the spinner 80 times: red 44, green 25, yellow 11. Find the observed probability of red.



d Percentage

Sefina tosses a coin 250 times: 135 heads and 115 tails. Find the observed probability of tails, as a percentage.



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER Observed probability $= \frac{\text{number of times it happened}}{\text{total number of trials}}$. Theoretical probability comes from equally likely outcomes; observed probability comes from results.

ROUND 1 Questions 1–4: find each observed probability: the number of times it happened ÷ the number of trials. Write it as a decimal.

1 A coin is tossed 250 times and lands heads 135 times. Find the observed probability of tails.

Working _____
P = _____

2 A die is rolled 80 times and a 1 or a 6 comes up 25 times. Find the observed probability of a 1 or a 6.

Working _____
P = _____

3 A goalkeeper faces 24 penalties and saves 18. Find the observed probability of a save.

Working _____
P = _____

4 Last year, 73 of the 365 days had rain. Find the observed probability of rain.

Working _____
P = _____

ROUND 2 Questions 5–8: decide whether each probability is theoretical (T) or observed (O). Write T or O, then give a reason.

5 40 of 100 Tweed residents surveyed own a dog, so P(owns a dog) = 0.4. T or O?

T or O _____
Reason _____

6 P(drawing an ace from a deck) $= \frac{1}{13}$. T or O?

T or O _____
Reason _____

7 60 of 200 cars passing the school were white, so P(white) = 0.3. T or O?

T or O _____
Reason _____

8 P(a random month starts with J) $= \frac{1}{4}$. T or O?

T or O _____
Reason _____

5 Observed probability

Probability • MA4-PRO-C-01

HARD •••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 11 QUESTIONS

REMEMBER Observed probability = frequency ÷ total trials. 'At least 50' means 50 or more. A biased die has outcomes that are not equally likely.

ROUND 3 Questions 9–11: the matches in each of 50 matchboxes were counted. Use the table.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

9 Find the observed probability that a box has at least 50 matches (50 or more), as a decimal.

Working

Answer

10 Find the observed probability that a box does not have exactly 50 matches, as a decimal.

Working

Answer

11 The label says 'average contents 50'. Find the mean number of matches per box. Is the label fair? Ch 7

Working

Answer

ROUND 4 Questions 12–14: a die was rolled 80 times. Use the table.

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

12 Find the observed probability of a 1, as a percentage.

Working

Answer

13 Find the observed probability of an even number, as a decimal.

Working

Answer

14 Do you think this die is biased? Explain using the numbers.

Working

Answer

ROUND 5 Questions 15–17: two dice were rolled 50 times and the sum was recorded each time. Use the table.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

15 Find the observed probability of a composite sum (a sum with more than two factors), as a decimal.

Working

Answer

16 Which sum or sums had an observed probability of $\frac{3}{25}$?

Working

Answer

17 Which sum had an observed probability of 16%?

Working

Answer

ROUND 6 Questions 18–19: run a simulation and watch the observed probability change.

18 Use RanInt#(1,2) (1 = heads) to toss a coin 40 times, in blocks of 10. Record the heads so far and the observed P(heads) after each block.

Tosses	10	20	30	40
Heads so far				
Observed P(heads)				

19 What happens to the observed P(heads) as the tosses increase? Combine your results with 4 classmates (200 tosses) and comment.

Answer

5 Observed probability

Probability • MA4-PRO-C-01

GOING FURTHER

WE DO with your teacher: copy every step into the space

Example 2 Use the tables. The total number of trials goes on the bottom of the fraction.

Matches in 50 boxes

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

A die rolled 80 times

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

a Table

Use the matchbox table. Find the observed probability that a box has exactly 51 matches.

b Fewer than

Use the matchbox table. Find the observed probability that a box has fewer than 50 matches.

c Biased?

Use the table of 80 rolls. Do you think the die is biased (unfair)? Give a reason using the numbers.

d Simulate

Use RanInt#(1,6) on a calculator to roll a die 20 times. Record the results, then find your observed probability of an even number.



✎ YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER Frequency = observed probability × number of trials. Use this to work backwards from a probability to a count.

1. After 50 tosses, the observed probability of heads is 0.58. After 30 more tosses, the observed probability for all 80 tosses is 0.55. How many heads were in the last 30 tosses?

Answer

2. A spinner is spun 120 times. The observed probability of red is 0.35. How many times did it land on red?

Answer

3. A die is rolled 40 times: 1 five times, 2 eight times, 3 x times, 4 six times, 5 seven times and 6 nine times. Find x and the observed P(3).

Answer

4. After 20 games, the observed probability that Kingscliff wins is 0.65. They win the next 5 games. Find the new observed probability of a win.

Answer

5. Use the dice-sum table from Round 5. Find the mean (average) sum. Ch 7

Answer

6. A biased coin lands tails 248 times in 400 tosses. Find the observed probability of heads.

Answer

✎ WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1. Jo's family simulation gave 4 girls 4 times in 50 trials. The theoretical P(4 girls) $= \frac{1}{16}$ (you will see why in Stage 5). Jo runs 50 more trials and gets 4 girls twice. Find the observed probability for all 100 trials. Is it closer to $\frac{1}{16}$?

Answer

2. A spinner has 3 colours. In 200 spins: red 98, blue 52, green 50. Sketch what the spinner probably looks like, and explain.

Answer

3. Two classes test a coin. 8K gets 23 heads in 40 tosses; 8R gets 112 heads in 200 tosses. Find each class's observed probability of heads. Which is the better estimate, and why?

Answer

6 Expected frequency

Probability • MA4-PRO-C-01

MEDIUM ●●

LEARNING INTENTION

We are learning to calculate the expected frequency of an event, and compare it with the observed frequency.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find expected frequency = probability × number of trials. ○○○
- *I can* compare observed and expected frequencies. ○○○
- *I can* use an observed probability to make a prediction. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Find each expected frequency: the probability × the number of trials.

a Coin

A coin is tossed 50 times. Find the expected number of heads.



b Die

A die is rolled 72 times. Find the expected number of 4s.



c Spinner

Vinh spins the spinner 80 times. Find the expected number of reds. He actually got 44 reds: compare.



d Rain

The probability of rain on an April day is 30%. April has 30 days. Find the expected number of rainy days.



YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER Expected frequency = probability × number of trials. For example, 6s in 60 rolls: $\frac{1}{6} \times 60 = 10$.

ROUND 1 Questions 1–4: find each expected frequency: probability × number of trials.

1 A die is rolled 300 times. How many 6s would you expect?



Probability _____
× number of trials _____
Expected _____

2 An 8-sided die has 2 red faces. It is rolled 200 times. How many reds would you expect?



Probability _____
× number of trials _____
Expected _____

3 A card is drawn from a deck and put back, 260 times. How many hearts would you expect?



Probability _____
× number of trials _____
Expected _____

4 The probability of rain each day is 0.35. How many rainy days would you expect in 20 days?



Probability _____
× number of trials _____
Expected _____

ROUND 2 Questions 5–8: Vinh's spinner (½ red, ¼ green, ¼ yellow) was spun 80 times: red 44, green 25, yellow 11. Compare expected with observed.

5 How many reds were expected? How many more or fewer were observed?

Working _____
Answer _____

6 How many greens were expected? How many more or fewer were observed?

Working _____
Answer _____

7 How many yellows were expected? How many more or fewer were observed?

Working _____
Answer _____

8 Do you think the spinner is fair? Explain using your answers.

Working _____
Answer _____

6 Expected frequency

Probability • MA4-PRO-C-01

HARD •••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER Expected = probability × trials. From data, use the observed probability. Small samples make poor predictions.

ROUND 3 Questions 9–11: a fair die is rolled 72 times. Find each expected frequency: probability × 72.

9 How many 4s would you expect?

Working _____

Answer _____

10 How many numbers greater than 2 would you expect?

Working _____

Answer _____

11 How many prime numbers would you expect?

Working _____

Answer _____

ROUND 4 Questions 12–14: use the matchbox sample of 50 boxes (table) to predict how many of 1000 boxes would hold each number of matches.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

12 How many of 1000 boxes would hold exactly 50 matches?

Working _____

Answer _____

13 How many would hold at least 50 matches (50 or more)?

Working _____

Answer _____

14 How many would hold fewer than 49 matches?

Working _____

Answer _____

ROUND 5 Questions 15–17: in 72 rolls of a die, a 4 came up 9 times. Predict the number of 4s in 1000 rolls.

15 Predict using the theoretical probability (to 1 decimal place).

Working _____

Answer _____

16 Predict using the observed probability.

Working _____

Answer _____

17 Which prediction do you trust more? Why?

Working _____

Answer _____

ROUND 6 Questions 18–20: two dice were rolled 50 times and the sum was recorded each time. Use the table.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

18 Using the observed probability, how many sums of 8 would you expect in 300 rolls?

Working _____

Answer _____

19 Using the observed probability, how many sums of 2 would you expect in 300 rolls? Why is this prediction unreliable?

Working _____

Answer _____

20 The theoretical probability of a sum of 7 is $\frac{1}{6}$ (Stage 5 shows why). How many 7s would you expect in 50 rolls? Compare with the table.

Working _____

Answer _____

WE DO with your teacher: copy every step into the space

Example 2 Expected = probability × trials. If the outcomes are not equally likely, use the observed probability from the data.

a From data

Harper's die gave 15 twos in 60 rolls. Use this observed probability to find how many 2s to expect in 500 rolls.



b Matchboxes

In a sample, 20 of 50 matchboxes held exactly 50 matches. How many of 1000 boxes would you expect to hold exactly 50?

c Not whole

A die is rolled 80 times. Find the expected number of 6s (to 1 decimal place). A 6 actually came up 22 times. What does this suggest?

d Lollies

For a bag of lollies, P(red) = 0.5, P(green) = 0.2 and P(purple) = 0.3. How many of each colour would you expect in a bag of 100 lollies?

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER Work backwards: probability = expected ÷ trials, and trials = expected ÷ probability.

1. The expected frequency of an event in 400 trials is 150. Find its probability.

Answer

2. A spinner is spun 250 times. The expected number of blues is 100. What fraction of the spinner is blue?

Answer

3. Among 80 cars, the expected number of white cars is 24. Find the probability that a car is not white.

Answer

4. A die is rolled n times. The expected number of 6s is 35. Find n, the number of rolls.

Answer

5. A spinner with 4 equal colour sectors was spun 200 times: red 52, blue 49, green 47, yellow 52. Find the expected number for each colour. Does the spinner seem fair?

Answer

6. In a bag of lollies, P(red) = 0.5 and P(green) = 0.2, and the rest are purple. The bag has 9 purple lollies. How many lollies would you expect in the whole bag?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1. A biased coin landed tails 248 times in 400 tosses. Use the observed probability to predict the number of heads in 1500 tosses.

Answer

2. 70 of the canteen's last 200 orders were wraps. Next week it expects 900 orders. How many wraps should it make? Why make a few extra?

Answer

3. In a game a die is rolled: Dani wins on 1 or 2, Van on 3, 4 or 5, and Kai on 6. They play 60 games. Find the expected number of wins for each player. Is the game fair?

Answer

E1 Venn diagrams

Extension • Stage 5 Path • MA5-PRO-P-01 — MEDIUM ●●

LEARNING INTENTION

We are learning to read and draw Venn diagrams, and use them to find probabilities with 'and', 'or' and 'not'.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

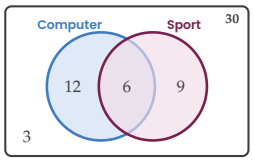
- *I can* read the four regions of a two-circle Venn diagram. ○○○
- *I can* draw a Venn diagram, starting with the overlap. ○○○
- *I can* tell inclusive 'or' from exclusive 'or' and find probabilities. ○○○

KEY IDEAS read these carefully, then highlight the key words — SELF-STUDY

KEY TERMS

Venn diagram: a rectangle (everyone) with a circle per group
A and B: in both groups (the overlap)
A or B (inclusive): in A, in B or in both
A or B, not both (exclusive): A only or B only
neither: outside both circles
mutually exclusive: can't happen together; no overlap

DIAGRAM



12 computer only, 6 both, 9 sport only, 3 neither

STEPS

1. start with the overlap (both)
2. A only = A total − both
3. B only = B total − both
4. neither = total − the other three
5. P = number in the region(s) ÷ total

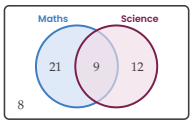
30 students: 18 computer, 15 sport, 6 both

WORKED worked examples: read each step, then cover it and try it yourself

Example 1 Use the Venn diagrams to answer each part. When you draw one, start with the overlap.

a Read

How many students were surveyed? How many like both subjects?



Add all four regions:
21 + 9 + 12 + 8 = **50** students
Both = the overlap = **9**

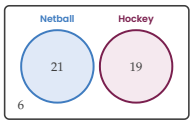
b Or

Same diagram. How many like Maths or Science? How many like Maths or Science, but not both?

Inside the circles: 21 + 9 + 12 = **42**
Outer parts only: 21 + 12 = **33**

c Mutually exclusive

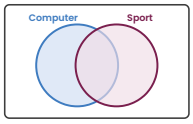
Are netball and hockey mutually exclusive? How many play netball or hockey?



No overlap, so **yes**: mutually exclusive.
Netball or hockey: 21 + 19 = **40**

d Draw

Of 30 students, 18 play computer games, 15 play sport and 6 do both. Fill in the diagram.

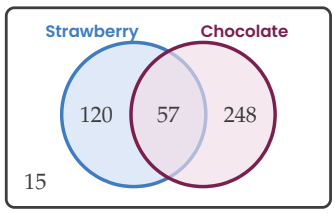


Overlap: **6**. Computer only: 18 − 6 = **12**. Sport only: 15 − 6 = **9**. Neither: 30 − 27 = **3**.

✎ YOU DO on your own: work down the box, one round at a time. Show your working. — 8 QUESTIONS

REMEMBER 'A and B' is the overlap; 'A or B' is inside the circles; 'not both' is the two outer parts; 'neither' is outside. (Not to scale.)

ROUND 1 Questions 1–4: people were asked which ice-cream flavours they like. Use the Venn diagram.



Each number is a count of people.

1 How many people were surveyed? (Add all four numbers.)

Answer _____

2 How many people liked strawberry or chocolate, but not both?

Answer _____

3 How many people liked neither flavour?

Answer _____

4 One person is picked at random. Find the probability that they like both flavours.

Answer _____

ROUND 2 Questions 5–8: mutually exclusive (cannot both happen) or overlapping? Explain.

5 Person: 'surfs' and 'is left-handed'. Mutually exclusive?

ME or overlap _____
Reason _____

6 Person: 'born in Australia' and 'born in Japan'. Mutually exclusive?

ME or overlap _____
Reason _____

7 Die: 'an even number' and 'a 3'. Mutually exclusive?

ME or overlap _____
Reason _____

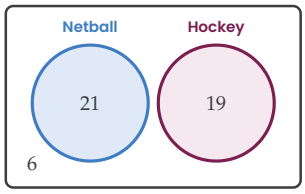
8 Die: 'an even number' and 'greater than 3'. Mutually exclusive?

ME or overlap _____
Reason _____

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER Start with the overlap. A only = A total − both. Neither = total − (A only + both + B only). Probability = number in the region ÷ total.

ROUND 3 Questions 9–11: this Venn diagram shows the sports some students play. Use it to answer.



9 How many students are in the diagram altogether?

Answer _____

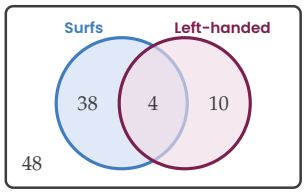
10 How many students play netball or hockey?

Answer _____

11 How many students play both netball and hockey?

Answer _____

ROUND 4 Questions 12–14: one person from this diagram is picked at random. Find each probability as a decimal.



12 Find the probability that the person surfs.

Answer _____

13 Find the probability that the person surfs or is left-handed.

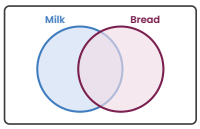
Answer _____

14 Find the probability that the person is left-handed and does not surf.

Answer _____

ROUND 5 Questions 15–17: a shop had 30 customers: 20 bought milk, 13 bought bread and 1 bought neither. Fill in the Venn diagram first, then answer.

15 How many customers bought both milk and bread? Write all four numbers in the diagram.



Answer _____

16 How many customers bought milk only?

Answer _____

17 One customer is picked at random. Find the probability they bought bread only.

Answer _____

ROUND 6 Questions 18–20: a group has 32 students with blue eyes, 38 with brown eyes and 10 with neither colour.

18 Are 'blue eyes' and 'brown eyes' mutually exclusive in this group? Explain.

Answer _____

19 How many students are in the group?

Answer _____

20 One student is picked at random. Find the probability that they do not have brown eyes, as a percentage.

Answer _____

WORKED worked examples: read each step, then cover it and try it yourself

Example 2 Find probabilities from Venn diagrams. The total number of people goes on the bottom of the fraction.

a And / not

Use Example 1d. One student is picked at random. Find P(plays sport but not computer games).

Sport but not computer = sport only = 9.

$P = \frac{9}{30} = \frac{3}{10}$

b Exclusive or

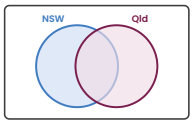
Use Example 1d. Find P(plays sport or computer games, but not both).

Not both = computer only + sport only = 12 + 9 = 21.

$P = \frac{21}{30} = \frac{7}{10}$

c Draw

Of 130 people, 44 like NSW holidays, 66 like Queensland and 20 like both. Fill in the diagram.



Overlap **20**. NSW only 44 − 20 = **24**. Qld only 66 − 20 = **46**. Neither 130 − 90 = **40**.

d Probability

Use part c. One person is picked at random. Find P(likes NSW only) and P(likes neither).

$P(\text{NSW only}) = \frac{24}{130} = \frac{12}{65}$

$P(\text{neither}) = \frac{40}{130} = \frac{4}{13}$

✎ YOU DO

on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER P(A or B) = P(A) + P(B) − P(A and B). Or draw the diagram and add the regions.

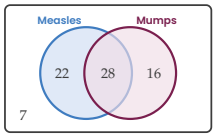
1 All 30 students in a class study IT or Chemistry (or both). 7 study both and 20 study IT. How many study Chemistry? Find the probability that a student picked at random studies Chemistry.

Answer

2 At camp, 40 students chose activities: 35 chose kayaking, 20 chose bushwalking, and everyone chose at least one. How many chose both? Find P(both) and P(not bushwalking).

Answer

3 The diagram shows how many children have had measles or mumps. One child is picked at random. Find P(measles), P(measles or mumps) and P(neither).



Answer

4 P(A) = 0.6, P(B) = 0.5 and P(A and B) = 0.3. Imagine 100 people: draw a Venn diagram with the number in each region, then find P(A or B).

Answer

5 In a Venn diagram, A only = x, both = 5, B only = 2x and neither = 4. The total is 39. Write an equation to find x, then find P(A).

Answer

6 P(A) = 0.7 and P(B) = 0.5. Can A and B be mutually exclusive? Explain.

Answer

✎ WORDED PROBLEMS

draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 Of 120 Year 8 students, 70 have a dog, 45 have a cat and 25 have neither. Draw a Venn diagram. Find P(a dog and a cat) and P(exactly one of the two).

Answer

2 Kingscliff Nippers has 48 members: 30 swim, 26 do board and 5 do neither. Draw a Venn diagram. Find the probability that a member picked at random swims but does not do board.

Answer

3 A number from 1 to 20 is chosen at random. A = 'even' and B = 'a multiple of 3'. Write the numbers 1 to 20 in a Venn diagram. Find P(A or B) and P(A or B but not both).

Answer

E2 Two-way tables

Extension • Stage 5 Path • MA5-PRO-P-01

MEDIUM ●●

LEARNING INTENTION

We are learning to read, complete and draw two-way tables, convert between Venn diagrams and two-way tables, and find probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* complete a two-way table using the totals. ○○○
- ▸ *I can* convert between a Venn diagram and a two-way table. ○○○
- ▸ *I can* find probabilities with 'and', 'or' and 'not' from a table. ○○○

KEY IDEAS

read these carefully, then highlight the key words — SELF-STUDY

KEY TERMS

two-way table: rows for one attribute, columns for another

each **cell** is an 'and', e.g. Year 7 *and* can swim

totals: row totals, column totals and the **grand total** (bottom right)

missing cell = total − the known cells

VENN DIAGRAM ↔ TABLE

	B	not B	Total
A	both	A only	A total
not A	B only	neither	
Total	B total		grand

overlap → 'A and B' cell; outside both circles → 'not A and not B' cell

STEPS

1. fill in every total you can
2. find missing cells by subtracting
3. check: rows and columns add up
4. P = cell(s) ÷ grand total

'or': add the cells, but count each cell once

WORKED

worked examples: read each step, then cover it and try it yourself

Example 1 The table shows which students can swim. Each cell is an 'and', e.g. Year 7 and can swim.

	Can swim	Cannot swim
Year 7	13	2
Year 8	9	3

(a) Total

How many students are in the table?

Add every cell:
13 + 2 + 9 + 3 = **27**

(b) Or

How many are in Year 7 or cannot swim (or both)?

All of Year 7 (13 + 2) plus Year 8 students who cannot swim (3):
15 + 3 = **18**
Don't count the 2 twice.

(c) And

How many are in Year 7 and cannot swim?

The cell in the Year 7 row and the Cannot swim column: **2**

(d) Probability

One student is picked at random. Find P(Year 8 and can swim).

Year 8 and can swim: 9 of 27.
$P = \frac{9}{27} = \frac{1}{3}$

✎ YOU DO

on your own: work down the box, one round at a time. Show your working. — 8 QUESTIONS

REMEMBER Each cell is an 'and'. For 'or', add the cells, counting each cell once. Probability = cell ÷ grand total.

ROUND 1

Questions 1–4: the players in a soccer club are grouped by age and weight. Use the table.

	Heavy	Light
Junior	64	96
Senior	144	32

1. How many players does the club have?

   Answer _____

2. How many players are juniors or light (or both)?

   Answer _____

3. How many players are juniors or light, but not both?

   Answer _____

4. One player is picked at random. Find the probability that they are a senior.

   Answer _____

ROUND 2

Questions 5–8: the table shows which students need glasses. Fill in the empty cells first, then answer.

	Needs glasses	Does not	Total
Year 9	22		82
Year 10		85	98
Total		145	

5. How many Year 9 students do not need glasses?

   Answer _____

6. How many Year 10 students need glasses?

   Answer _____

7. How many students are there in total?

   Answer _____

8. One student is picked at random. Find the probability that they need glasses.

   Answer _____

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER Missing cell = total − the known cells. The Venn overlap goes in the 'A and B' cell; outside both circles goes in the 'not A and not B' cell.

ROUND 3 Questions 9–11: the table shows who likes broccoli. Fill in the empty cells first, then answer.

	Likes	Dislikes	Total
Year 7	60		150
Year 8		40	
Total	130		

9 How many students dislike broccoli?

Answer _____

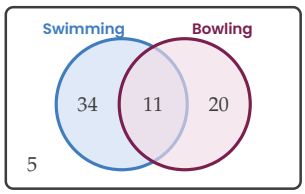
10 One student is picked at random. Find the probability that they are in Year 7 or dislike broccoli.

Answer _____

11 Find the probability that the student is in Year 8 and likes broccoli.

Answer _____

ROUND 4 Questions 12–14: turn the Venn diagram into a two-way table, then answer.



12 Fill in the table from the Venn diagram (S = swims, B = bowls).

	S	not S	Tot
B			
not B			
Tot			

13 One person is picked at random. Find the probability that they swim.

Answer _____

14 Find the probability that the person bowls but does not swim.

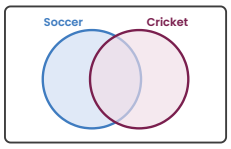
Answer _____

ROUND 5 Questions 15–17: turn the two-way table into a Venn diagram, then answer.

	Cricket	Not cricket
Soccer	28	15
Not soccer	20	7

TV sports watched by 70 people

15 Write the four numbers from the table in the Venn diagram.



16 One person is picked at random. Find the probability that they watch soccer or cricket, but not both.

Answer _____

17 Find the probability that the person does not watch cricket.

Answer _____

ROUND 6 Questions 18–20: a sports high school surveyed students about hockey and judo. Fill in the table first, then answer.

	Judo	Not judo	Total
Hockey	128		360
Not hockey	100	40	
Total			

18 One student is picked at random. Find the probability that they do judo.

Answer _____

19 Find the probability that the student does judo but not hockey.

Answer _____

20 Find the probability that the student does hockey or judo, but not both.

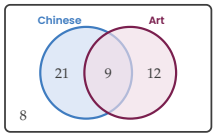
Answer _____

WORKED worked examples: read each step, then cover it and try it yourself

Example 2 **Complete the cinema table, use it to find probabilities, then turn the Venn diagram into a table.**

	Under 18	18 and over	Total
Saturday		142	198
Sunday	45		
Total			344

A cinema's audience on two days



a Complete

Fill in the empty cells. Each row and column must add to its total.

198 − 142 = **56**; 344 − 198 = **146** (Sunday)
146 − 45 = **101**
56 + 45 = **101**; 142 + 101 = **243**

b And

One person is picked at random. Find P(Sunday and 18 or over).

Sunday and 18+: 101 of 344.
$P = \frac{101}{344}$

c Or

Find P(Sunday or 18 or over), counting each person once.

All of Sunday (146) plus Saturday 18+ (142): 288.
$P = \frac{288}{344} = \frac{36}{43}$

d Convert

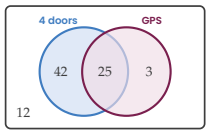
Show the Venn diagram (50 students) as a two-way table: Art / not Art down the side.

	Ch	not	Tot
Art	9	12	21
not	21	8	29
Tot	30	20	50

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER Draw the table or the diagram first. Every row and column must add to its total.

1 The Venn diagram shows the cars at Rusty Motors. Draw it as a two-way table. Then find the probability that a car picked at random has GPS.



Answer

2 Use your table from Question 1. Find P(4 doors and GPS) and P(no GPS).

Answer

3 A two-way table has a grand total of 80. The A total is 30, the B total is 45 and the 'A and B' cell is 12. Find the 'not A and not B' cell.

Answer

4 8K has 30 students: 18 walk to school, and 9 of the walkers are in the band. 4 students who don't walk are in the band. Make a two-way table. Find the probability that a student picked at random is in the band.

Answer

5 In Year 8, 40% surf, 25% skate and 10% do both. Find the probability that a student does neither, as a percentage.

Answer

6 P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Make a two-way table of probabilities (the grand total is 1) and find P(neither).

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1 The canteen had 200 orders: 120 included a drink, 90 were hot food, and 50 were hot food with a drink. Make a two-way table. Find the probability that an order was hot food or included a drink, but not both.

Answer

2 Of 150 Tweed residents, 40 own a boat, 55 live on the river and 18 do both. Find P(owns a boat but does not live on the river) and P(neither).

Answer

3 Explain why a two-way table can show Years 7, 8 and 9 (down the side) more easily than a Venn diagram.

Answer

CHAPTER SUMMARY

the key facts from every lesson, and how confident you are with each

HOW CONFIDENT AM I?

○○○ not yet · getting there · got it

1	Chance and the sample space	○○○	2	Probability of an event	○○○
3	Complementary events	○○○	4	Probability problems	○○○
5	Observed probability	○○○	6	Expected frequency	○○○
E1	Venn diagrams	○○○	E2	Two-way tables	○○○

Key facts

Sample space: every possible outcome, each written once, in braces { } (L1)

P(E) $= \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ when outcomes are equally likely (L2)

'or' for different groups: add the counts, e.g. P(red or blue) (L2)

Checklist: total → favourable → 'not'? → form asked → check (L4)

More trials: the observed probability usually gets closer to the theoretical one (L5)

Fair or biased? Close to expected: probably fair. Far off over many trials: probably biased (L6)

Two-way table: each cell is an 'and'; missing cell = total − known cells; P = cell ÷ grand total (E2)

Probability scale: 0 impossible · $\frac{1}{2}$ even chance · 1 certain (L1)

Converting: fraction → decimal: top ÷ bottom; decimal → percentage: × 100% (L2)

Complement: P(not E) = 1 − P(E); all probabilities add to 1 (L3)

Observed P(E) $= \frac{\text{number of times E happened}}{\text{total number of trials}}$ (L5)

Expected frequency = probability × number of trials (L6)

Venn diagram: A and B = the overlap; A or B = inside either circle; not both = the outer parts; mutually exclusive = no overlap (E1)

IF YOU WERE AWAY

two examples from each lesson, fully worked: read them, then try the lesson's first round

Lesson 1 • Chance and the sample space

the lesson is on p. 2

Example 1: a sample space

List the sample space for a spinner with sectors red, blue, red, green.

Step 1 Write each colour once: {red, blue, green}.

Step 2 Count: 3 outcomes.

Step 3 Red has 2 of the 4 sectors, so the outcomes are **not equally likely**.

Example 2: chance words

The forecast says a 30% chance of rain. Describe the chance.

Step 1 Compare with 50% (even chance): 30% is less.

Step 2 So rain is **unlikely**.

Step 3 It is not 0%, so it is still possible.

Lesson 2 • Probability of an event

the lesson is on p. 5

Example 1: P(E) from counters

A bag has 5 red, 3 blue and 4 green counters. Find P(red).

Step 1 Total = 5 + 3 + 4 = 12.

Step 2 Favourable: 5 red.

Step 3 P(red) $= \frac{5}{12}$

Example 2: a percentage

A die is rolled. Find P(a number greater than 4), as a percentage.

Step 1 Favourable: {5, 6}, 2 of 6.

Step 2 $\frac{2}{6} = \frac{1}{3}$

Step 3 $\frac{1}{3} \times 100\% = 33\frac{1}{3}\% \approx 33.3\%$

Lesson 3 • Complementary events

the lesson is on p. 8

Example 1: P(not E) with decimals

$P(\text{rain}) = 0.15$. Find $P(\text{no rain})$.

Step 1 'No rain' is the complement of 'rain'.

Step 2 $\text{P}(\text{no rain}) = 1 - 0.15 = \mathbf{0.85}$

Example 2: P(not E) with fractions

A pack has 10 red, 6 yellow and 4 green cards. Find $P(\text{not yellow})$.

Step 1 $\text{P}(\text{yellow}) = \frac{6}{20} = \frac{3}{10}$

Step 2 $\text{P}(\text{not yellow}) = 1 - \frac{3}{10} = \mathbf{\frac{7}{10}}$

Lesson 4 • Probability problems

the lesson is on p. 11

Example 1: an 'or' problem

A number from 1 to 15 is chosen. Find $P(\text{a multiple of 3 or 5})$.

Step 1 List: $\{3, 5, 6, 9, 10, 12, 15\}$ (15 once).

Step 2 7 of 15: $\text{P} = \mathbf{\frac{7}{15}}$

Example 2: no memory

A coin has shown tails 7 times in a row. Find $P(\text{tails next})$.

Each toss is a fresh start: the coin does not remember.

$\text{P}(\text{tails}) = \mathbf{\frac{1}{2}}$

Lesson 5 • Observed probability

the lesson is on p. 14

Example 1: observed probability

A die is rolled 60 times; 2 comes up 15 times. Find the observed $P(2)$.

Step 1 Observed P = times ÷ trials $= \frac{15}{60} = \mathbf{\frac{1}{4}}$

Step 2 Theoretical $\text{P}(2) = \frac{1}{6}$, so 2 came up more than expected.

Example 2: from a table

Matches in 50 boxes: 48 (3), 49 (6), 50 (20), 51 (16), 52 (4), 53 (1). Find the observed $P(\text{fewer than } 50)$.

Step 1 Fewer than 50: 48 or 49, $3 + 6 = 9$ boxes.

Step 2 $\text{P} = \mathbf{\frac{9}{50}}$

Lesson 6 • Expected frequency

the lesson is on p. 17

Example 1: expected frequency

A die is rolled 72 times. Find the expected number of 4s.

Expected = P × trials $= \frac{1}{6} \times 72 = \mathbf{12}$

Example 2: predicting from data

Harper's die gave 15 twos in 60 rolls. How many 2s in 500 rolls?

Step 1 Observed $\text{P}(2) = \frac{15}{60} = \frac{1}{4}$

Step 2 $\frac{1}{4} \times 500 = \mathbf{125}$

Lesson E1 • Venn diagrams

the lesson is on p. 20

Example 1: drawing a Venn diagram

30 students: 18 computer, 15 sport, 6 both.

Step 1 Overlap: 6.

Step 2 Computer only $18 - 6 = 12$; sport only $15 - 6 = 9$.

Step 3 Neither: $30 - 27 = 3$.

Example 2: exclusive or

Same diagram: find P(sport or computer, but not both).

Not both = the two outer parts: $12 + 9 = 21$.

$\mathrm{P} = \frac{21}{30} = \mathbf{\frac{7}{10}}$

Lesson E2 • Two-way tables

the lesson is on p. 23

Example 1: reading a two-way table

Year 7: 13 can swim, 2 cannot; Year 8: 9 can, 3 cannot. Find P(Year 8 and can swim).

Step 1 Grand total 27.

Step 2 The Year 8 / can swim cell: 9.

Step 3 $\mathrm{P} = \frac{9}{27} = \mathbf{\frac{1}{3}}$

Example 2: a missing cell

Saturday: under 18 ?, 18 and over 142, total 198. Find the missing cell.

missing = total − known = $198 - 142 = \mathbf{56}$

R1 [L1] A die is rolled. Put these in order from least to most likely: A a 1, B an odd number, C at most 5.

R2 [L2] An Esky has 6 lemonade and 4 lime cans. How many colas must be added so that $P(\text{cola}) = \frac{1}{3}$?

R3 [L2] One card is taken from a deck. Find the probability of a red card or a king (count each card once).

R4 [L2] A spinner has sector angles 120° red, 150° blue and 90° green. Find P(blue).

R5 [L3] The probability that E does not happen is 0.62. Find P(E).

R6 [L3] A spinner is red, green or blue. $P(\text{red}) = 2 \times P(\text{green})$ and $P(\text{blue}) = 0.4$. Find P(green).

R7 [L3] A 3-digit number (100 to 999) is chosen at random. Find P(not a multiple of 5).

R8 [L4] A whole number from 1 to 30 is chosen at random. Find the probability of a multiple of 3 or 5.

R9 [L4] A die is weighted so $P(6) = 0.4$ and the other numbers are equally likely. Find P(2).

R10 [L5] After 40 games, the observed P(win) = 0.6. The team then loses 10 games in a row. Find the new observed P(win).

R11 [L5] A die is rolled 120 times: 1 (18), 2 (22), 3 (20), 4 (19), 5 (21), 6 (x). Find x and the observed P(6).

R12 [L6] The expected frequency of an event in 400 trials is 140. Find its probability.

R13 [L6] The observed probability that a car is white is 0.3. How many white cars would you expect among 450 cars?

R14 [L6] A 4-sector spinner (equal sectors) gave 160 reds in 400 spins. Is it fair? Use the expected frequency.

R15 [E1] 60 students: 35 play sport, 28 play an instrument and 12 do both. How many do neither? Find P(exactly one).

R16 [E1] On a die, are 'a prime number' and 'an even number' mutually exclusive? Explain.

R17 [E1] P(A) = 0.45, P(B) = 0.3 and P(A and B) = 0.1. Find P(neither).

R18 [E2] A two-way table: Year 7 has 40 like and 70 in total; Year 8 has 25 dislike; the grand total is 130. Find P(Year 8 and likes).

R19 [E2] A Venn diagram has A only 14, both 6, B only 10 and neither 20. Make a two-way table and find P(not A).

R20 [E2] Use the previous question. Find P(A or B, but not both).

EXIT TICKET Lesson 1: Chance and the sample space Name _____ Class _____

Medium ●● 2 coins are tossed. List the possible numbers of heads (the sample space).

Hard ●●● Are those numbers of heads equally likely? Explain.

Going further Design a spinner with 6 equal sectors where red is likely and blue is unlikely but possible.

EXIT TICKET Lesson 2: Probability of an event Name _____ Class _____

Medium ●● One card is taken from a deck. Find the probability of a picture card (J, Q or K).

Hard ●●● A 10-sided die (1 to 10) is rolled. Find the probability of a multiple of 4, as a decimal.

Going further A bag has 4 red counters and some blue ones. P(red) = 0.2. How many blue counters are there?

EXIT TICKET Lesson 3: Complementary events Name _____ Class _____

Medium ●● A die is rolled. Write the complement of 'at least 3', then find its probability.

Hard ●●● A spinner is red, blue or green. P(red) = 0.3 and P(blue) = 0.45. Find P(not green).

Going further The probability that E does not happen is four times P(E). Find P(E).

EXIT TICKET Lesson 4: Probability problems Name _____ Class _____

Medium ●● A whole number from 1 to 100 is chosen at random. Find the probability of a square number.

Hard ●●● A whole number from 1 to 20 is chosen at random. Find the probability of a multiple of 2 or 5.

Going further A die is rolled: Amy wins on 1 or 2, Bo wins on 3 to 6. Is the game fair? Explain.

✂

SUMMARISE LESSON 1 IN YOUR OWN WORDS

Chance and the sample space • the back of your exit ticket

✂

SUMMARISE LESSON 2 IN YOUR OWN WORDS

Probability of an event • the back of your exit ticket

✂

SUMMARISE LESSON 3 IN YOUR OWN WORDS

Complementary events • the back of your exit ticket

✂

SUMMARISE LESSON 4 IN YOUR OWN WORDS

Probability problems • the back of your exit ticket

✂

EXIT TICKET Lesson 5: Observed probability Name _____ Class _____

Medium ●● A coin is tossed 400 times and lands tails 150 times. Find the observed probability of tails, as a decimal.

Hard ●●● After 20 shots, a player's observed probability of a goal is 0.45. How many goals has the player scored?

Going further The player then scores with all of the next 5 shots. Find the new observed probability of a goal.

EXIT TICKET Lesson 6: Expected frequency Name _____ Class _____

Medium ●● A spinner is ¼ red. It is spun 200 times. How many reds would you expect?

Hard ●●● An event is expected 45 times in 180 trials. Find its probability.

Going further An event has an observed probability of 0.35. How many times would you expect it in 500 trials?

EXIT TICKET Lesson E1: Venn diagrams Name _____ Class _____

Medium ●● A Venn diagram has A only 8, both 3, B only 5 and neither 4. Find P(A or B).

Hard ●●● Use the same Venn diagram. Find P(A or B, but not both).

Going further Of 40 students, 22 surf, 15 skate and 7 do neither. How many students do both?

EXIT TICKET Lesson E2: Two-way tables Name _____ Class _____

Medium ●● A two-way table: Year 7 has 12 like, 8 dislike; Year 8 has 15 like, 5 dislike. Find P(Year 7 and likes).

Hard ●●● Use the same table. Find P(Year 8 or dislikes).

Going further Draw a Venn diagram for that table, with one circle for Year 7 and one for likes.

✂

SUMMARISE LESSON 5 IN YOUR OWN WORDS

Observed probability • the back of your exit ticket

✂

SUMMARISE LESSON 6 IN YOUR OWN WORDS

Expected frequency • the back of your exit ticket

✂

SUMMARISE LESSON E1 IN YOUR OWN WORDS

Venn diagrams • the back of your exit ticket

✂

SUMMARISE LESSON E2 IN YOUR OWN WORDS

Two-way tables • the back of your exit ticket

✂

HOMEWORK STRIPS — after each lesson, cut out its strip (start at the top right), do it at home, and hand it in next lesson

HW 7 E1 Venn ✂

1 A Venn diagram has A only 9, both 4, B only 7 and neither 5. Find P(A).

2 Use the same Venn diagram. Find P(A or B).

3 Use the same Venn diagram. Find P(exactly one of A and B).

HW 4 L4 Problems ✂

1 A whole number from 1 to 30 is chosen at random. Find the probability of a multiple of 4 or 6.

2 In a group, P(left-handed) = 0.12 and 6 students are left-handed. How many students are in the group?

3 A die is rolled: Ann wins on a prime number, Bo on any other number. Is the game fair? Explain.

HW 1 L1 Chance ✂

1 3 coins are tossed. List the possible numbers of heads.

2 Are those numbers of heads equally likely? Explain.

3 Put these probabilities in order from least to most likely: 0.3, $\frac{2}{5}$, 35%.

HW 8 E2 Tables ✂

1 A two-way table has: A and B 6, A not B 9, B not A 5, neither 10. Find P(B).

2 Use the same table. Find P(A or B).

3 Use the same table. Find P(not A and not B).

HW 5 L5 Observed ✂

1 A coin is tossed 250 times and lands heads 140 times. Find the observed probability of heads.

2 After 30 spins, the observed probability of red is 0.4. The next 10 spins are all red. Find the new observed probability of red.

3 Why do two classes get different observed probabilities?

HW 2 L2 P(E) ✂

1 One card is taken from a deck. Find the probability of a red picture card.

2 An Esky has 6 lemonade and 4 lime cans. How many cola cans must be added so that P(lemonade) = $\frac{1}{2}$?

3 A spinner has a red sector with an angle of 90°. Find the probability of red.

HW 9 Before the test ✂

1 **L3** One tile is picked from KINGSCLIFF. Find the probability that it is not a vowel.

2 **L6** A die is rolled 150 times. How many 4s would you expect?

3 **E1** On a die, are ‘a multiple of 3’ and ‘a 5’ mutually exclusive?

HW 6 L6 Expected ✂

1 An event E is expected 90 times in 360 trials. Find P(E).

2 A die is rolled n times. The expected number of 6s is 42. Find n.

3 An event has an observed probability of 0.15. How many times would you expect it in 2000 trials?

HW 3 L3 Complements ✂

1 The probability that E does not happen is 0.35. Find P(E).

2 A spinner is red, blue or green. P(red) = 0.25 and P(blue) = 0.4. Find P(green).

3 A whole number from 1 to 50 is chosen at random. Find the probability that it is not a multiple of 7.

HW 1 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 2 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 3 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 4 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 5 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 6 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 7 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 8 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 9 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working: